

12.1.0

ELECTRICAL ENGINEERING PRINCIPLES

12.1.01

Introduction

The module unit is intended to equip the trainee with knowledge, skills and attitude required to apply the basic electrical engineering principles necessary for training in the area of trade.

12.1.02

General Objectives

By the end of the module unit, the trainee should be able to:

- understand the concepts of electrical quantities
- perform basic calculations on dc and ac circuits
- explain the behaviour of magnetic materials, magnetism and electromagnetism
- describe the construction and operation of dc sources
- explain the basic concepts of ac generation
- describe the theory of dielectrics, principles and characteristics of capacitors
- analyze analogue ac and dc meters

12.1.03

Module Unit Summary and Time Allocation**Electrical Engineering Principles**

Code	Sub Module Unit	Content	Time Hrs		
			Theory	Practice	Total
12.1.1	Electrical quantities and application	- Basic SI units - Derived units - Units of electrical quantities - Calculations	4	4	8
12.1.2	Cells and Batteries	- Electrolysis and applications - Definition of electricity - Construction of simple cell - Determination of E.m.f in cells - Differentiation of	4	2	6

		primary and secondary • Construction and applications of cells • Calculations • Charging methods			
12.1.3	Electrical Instruments	• Analogue instruments • Repulsion and attraction moving coil instruments • Shunts and multipliers • Multimeters	4	6	10
12.1.4	DC Circuit Theory	• Definition of resistivity • Ohms law • Series circuits • Parallel circuits • Series-parallel networks • Kirchhoff's Laws, • Superposition theorem • Calculations • Chemical effects due to electric current	6	2	8
12.1.5	Magnetism & Electromagnetism	• Electromagnetic fields • Leakage and flux fringing • Laws of Electromagnetic induction • Inductance • Hysteresis • Calculations	6	2	8
12.1.6	Electrostatics	• Definitions • Types of capacitors • Concept of charge and electrostatic field • Dielectric field • Calculations	6	2	8
12.1.7	AC Theory	• Simple ac generator • Terminologies applied to ac generators	6	2	8

		• Unidirectional and directional waveforms • Passive elements in ac circuits • Calculations			
12.1.8	Single Phase Transformers	• Constructional features • Principles of operation • Tests for single phase transformers • Applications of single phase transformers	8	2	10
Total Time			44	22	66

12.1.1 ELECTRICAL QUANTITIES AND APPLICATION

Theory

12.1.1T0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) state the basic SI units.
- b) recognize derived SI units.
- c) state various units of electrical quantities
- d) perform simple calculations

Content

12.1.1T1 Basic SI unit quantities

- i) Length
- ii) Mass
- iii) Time

12.1.1T2 Derived SI units

12.1.1T3 Electrical quantities

- i) Coulomb
- ii) Newton
- iii) Joule
- iv) Volt
- v) Ohm
- vi) Siemens
- vii) Watt

12.1.1T4 Simple calculations

12.1.2 CELLS AND BATTERIES

Theory

12.1.2T0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) describe electrolysis and its applications

- b) define the term electricity
- c) describe the construction and operation of a simple cell
- d) determine the total e.m.f and internal resistance of cells connected in series and in parallel
- e) distinguish between primary and secondary cells
- f) explain the applications and construction of different types of cells
- g) perform calculations on secondary cells
- h) describe charging methods of batteries

Content

12.1.2T1 Description of electrolysis and its applications

12.1.2T2 Definition of electricity

12.1.2T3 Construction and operation of a simple cell

12.1.2T4 Determination of total internal resistance of cells connected in series and in parallel, and hence the E.m.f

12.1.2T5 Distinguishing between primary and secondary cells

12.1.2T6 Construction and applications of different types of cells

12.1.2T7 Calculations

- i) E.m.f
- ii) Internal resistance

12.1.2T8 Charging methods

- i) Constant current
- ii) Constant voltage
- iii) Trickle charge
- iv) Booster charge
- v) Battery ratings

vi) Simple calculations

Practice

- 12.1.2P0 *Specific Objectives*
By the end of the sub module unit, the trainee should be able to:
- perform experiments on electrolysis
 - determine internal resistance of cells connected in series and in parallel
 - perform experiments on charging of batteries

12.1.2C Competence

The trainee should have the ability to:

- Perform experiments on electrolysis
- Determine internal resistance of cells connected in series and in parallel
- Charge batteries

Content

- 12.1.2P1 Electrolysis
12.1.2P2 Internal resistance of cells connected in series and in parallel
12.1.2P3 Battery charging

Suggested Learning/Teaching Resources

- Various batteries
- Sulphuric acid
- Distilled water
- Battery chargers
- Test instruments

12.1.3 ELECTRICAL INSTRUMENTS

Theory

- 12.1.3T0 *Specific Objectives*
By the end of the sub module unit, the trainee should be able to:

- describe the construction and operation of electrical instruments
- describe the construction and operation of the thermocouple
- extend the range of meters using shunts and multipliers
- explain the operation of a multimeter

12.1.3C Competence

The trainee should have the ability to:

- identify various measuring instruments
- perform electrical measurements using the instruments
- repair and maintain electrical measuring instruments

Content

- 12.1.3T1 Construction and operation of electrical instruments
- Moving iron
 - Moving coil

12.1.3T2 Thermocouple

- i) Advantages and disadvantages
- ii) Applications

12.1.3T3 Extension of range of instruments

- i) Shunts
- ii) Multipliers
- iii) Operation of multimeter
- iv) Ammeter
- v) Voltmeter
- vi) Ohmmeter

Practice

12.1.3P0 Specific Objectives

By the end of the sub module unit, the trainee should be able to:

- a) identify different types of instruments
- b) perform measurements using instruments
- c) repair and maintain instruments

Content

12.1.3P1 Types of instruments (analogue/digital)

- i) Voltmeters
- ii) Ammeters
- iii) Ohmmeters
- iv) Frequency meters
- v) Wattmeter
- vi) Multimeters
- vii) Cathode – Ray

Oscilloscope

12.1.3P2 Measurements of electrical quantities

- i) Voltage or potential drop

ii) Current

iii) Resistance or continuity

iv) Frequency

v) Power

vi) Time and period of waveforms

vii) Testing components

12.1.3T3 Repair and Maintenance of instruments

i) Voltmeters

ii) Ammeters

iii) Ohmmeters

iv) Frequency meters

v) Wattmeter's

vi) Multimeters

vii) Cathode – ray

oscilloscope

viii) Safe use of instruments

Suggested Learning/Teaching Resources

i) Voltmeters

ii) Ammeters

iii) Ohmmeters

iv) Frequency meters

v) Wattmeter's

vi) Multimeters

vii) Cathode – ray

oscilloscope

viii) Conductors

12.1.4 DC CIRCUIT THEORY

Theory

12.1.4T0 Specific Objectives

By the end of the sub module unit, the trainee should be able to:

- a) define resistivity

- b) define Ohm's
- c) calculate voltage, current and resistance in a series circuit
- d) calculate voltage, current and resistance in a parallel circuit
- e) calculate voltage, current and resistance in a series-parallel circuit
- f) state Kirchoff's laws
- g) state the superposition theorem
- h) state Norton's theorem
- i) state Thevenin's theorem
- j) Solve simple circuit problems using Kirchoff's, superposition, Norton's and Thevenin's theorems
- k) explain chemical effects due to electric current

- 12.1.4C Competence**
The trainee should have the ability to:
- i) Measure electrical quantities
 - ii) Determine conductor resistance

Content

- 12.1.4T1 Definition of:**
- i) Resistivity
 - ii) Conductors
 - iii) Conductivity

- iv) Effect of temperature change in the value of resistance
- v) Effect of positive temperature coefficient
- vi) Alloyed resistance temperature coefficient
- vii) Heat energy released by current

- 12.1.4T2 Ohm's Law
- 12.1.4T3 Voltage, current and resistance in a series circuit
- 12.1.4T4 Voltage, current and resistance in a parallel circuit
- 12.1.4T5 Voltage, current and resistance in a Series-parallel circuit
- 12.1.4T6 Statement of Kirchoff's theorem
- 12.1.4T7 Statement of Superposition's theorem
- 12.1.4T8 Statement of Norton's theorems
- 12.1.4T9 Statement of Thevenin's theorems
- 12.1.4T10 Superposition theorem
- 12.1.4T11 Solving circuit problems using Kirchoff's and Superposition theorems
- 12.1.4T12 Chemical effects due to electric current
 - i) Electrolysis
 - ii) Faraday's Laws of electrolysis
 - iii) Practical use of electrolytic action
 - iv) Electro-plating

- v) Electrolytic corrosion
- vi) Practice

12.1.4T4 Experiments on charging and discharging of capacitors

12.1.4P0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) connect simple electrical circuits
- b) measure various electrical quantities
- c) verify Ohm's law
- d) verify that the resistance of a material depends on area, length and resistivity
- e) verify Kirchhoff's law

Suggested Learning/Teaching Resources

- Dc power source
- Assorted resistance
- Measuring instruments
- Bread boards
- Connecting leads

12.1.5 MAGNETISM AND ELECTROMAGNETISM

Theory

12.1.5T0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) describe the concept of magnetic fields and field distribution
- b) explain leakage flux and fringing
- c) describe the basic laws of electromagnetic induction
- d) explain the concepts of inductance
- e) plot and explain the hysteresis loop
- f) explain and solve problems relating to magnetic circuits

Content

- 12.1.4P1 Simple electrical circuits
- 12.1.4P2 Measurement of electrical quantities
 - i) Current
 - ii) Voltage
 - iii) Resistance
 - iv) Power
- 12.1.4T3 Verification of Ohm's law
- 12.1.4T4 Determination of conductor resistance
 - i) Resistance
 - ii) Resistivity
 - iii) length
 - iv) Area
- 12.1.4T5 Verification of kirchhoff's laws
 - i) Current law
 - ii) Voltage law
- 12.1.4T3 Testing of capacitance

12.1.5C Competence

The trainee should have the ability to:

- i) Construct an electromagnet
- ii) Apply magnets in the engineering field

Content

12.1.5T1 Description of electromagnetic fields

- i) Permanent magnet
- ii) Between a pair of poles
- iii) In a current carrying conductor

iv) In parallel wires

v) In a loop

vi) In a solenoid

12.1.5T2 Leakage flux and fringing

12.1.5T3 Laws of electromagnetic induction

- i) Faraday's law
- ii) Lenz's law
- iii) Fleming's rules

12.1.5T4 Inductance

- i) Self
- ii) Mutual
- iii) Calculations
- iv) Practical applications of induced electromotive force

12.1.5T5 Hysteresis loop

- i) Plotting
- ii) Explanation

12.1.5T6 Problems relating to magnetic circuits

- i) Series
- ii) parallel

Practice

12.1.5P0 Specific Objectives

By the end of the sub module unit, the trainee should be able to:

- a) identify various types of magnetic materials
- b) verify the existence of magnetic field
- c) apply the principles of electromagnetism

Content

12.1.5P1 Magnetic materials

- i) Identification
- ii) Classification
- iii) Magnetic
- iv) Non magnetic

12.1.5P2 Verification of the existence of magnetic field

- i) Bar magnet
- ii) Horse shoe magnet
- iii) Combination of magnets
- iv) Current carrying conductor

12.1.5P3 Applications of electromagnetism

- Construction of electromagnet

Suggested Learning/Teaching Resources

- Permanent magnets
- Electromagnet
- Power
- Wires
- Bells
- Soft iron

12.1.6 ELECTROSTATICS

Theory

12.1.6T0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) define electrostatic quantities
- b) types of capacitors
- c) describe the concept of charge and electrostatic field
- d) explain the dielectric effect
- e) perform calculations involving capacitance

12.1.6C **Competence**

The trainee should have the ability to:

- i) Identify capacitors
- ii) Measure capacitance
- iii) Test capacitance
- iv) Apply capacitors in electrical circuits

Content

12.1.6T1 Definitions of electrostatic quantities and units

- i) Electric flux
- ii) Electric flux density
- iii) Electric field intensity
- iv) Permittivity
- v) Capacitance
- vi) Charge
- vii) Derivation of the capacitance formula

12.1.6T2 Types of capacitors

- i) Paper capacitors
- ii) Electrolytic capacitors
- iii) Ceramic capacitors
- iv) Aluminium foil capacitor
- v) Polyester capacitor
- vi) Tantalum capacitor
- vii) Multiplate capacitor
- viii) Variable capacitor
- ix) Applications
- x) Description of charge concept

12.1.6T3 Charge

- i) Electric field strength
- ii) Capacitance and its units

12.1.6T4 Dielectric field effect

12.1.6T5 Calculations

- i) Capacitors in series
- ii) Capacitors in parallel
- iii) Energy stored in a capacitor

Practice

12.1.6P0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) identify various types of
- b) capacitors
- c) measure capacitance in
- d) connections
- e) test a capacitor
- f) perform experiments on charging and discharging of capacitors

Content

12.1.6P1 Types of capacitors

12.1.6T2 Measurement of capacitance in various connections

Testing of capacitance

12.1.6T3 Experiments on charging and discharging of Capacitors

ii) Install power factor correction equipment

Suggested Teaching/Learning Resources

- Assorted capacitors
- Test instruments
- Bread boards

12.1.7 AC THEORY

Theory

12.1.7T0 *Specific Objectives*

By the end of the sub module unit, the trainee should be able to:

- a) describe the principle of ac generators
- b) explain the terms used with ac generators
- c) derive the e.m.f equation
- d) explain the effects of various passive elements in an ac circuits
- e) perform calculations related to sinusoidal waveforms

12.1.7C **Competence**

The trainee should have the ability to:

- i) Perform power factor improvement

12.1.7T1 *Content*
Description of principles of AC generation

i) Effect of rotating a coil in a magnetic field

ii) Waveforms for unidirectional and alternating fields

iii) Graphical illustrations of AC waveforms

12.1.7T2 Terms used in generators

i) Waveforms

ii) Period

iii) Frequency

iv) Instantaneous value

v) Amplitude, average values

vi) Root mean square values and its derivation by graphical methods

vii) Graphical addition of two sinusoidal waveforms

viii) Effects of passive elements in ac circuits

- Resistance
- Capacitance
- Inductance
- Phase difference
- Vector diagrams

12.1.7T3 Derivation of the EMF equation

- 12.1.7T4 Effects of passive elements in an ac circuit
12.1.7T5 Calculations

Practice

- 12.1.7P0 *Specific Objectives*
By the end of the sub module unit, the trainee should be able to:
- identify components of an ac generator
 - generate a sine wave
 - establish the features of an ac waveform
 - verify the effects of passive elements in ac circuits
 - perform experiments to show the effect of power factor

Content

- 12.1.7P1 Identification of components of ac generator
12.1.7T2 Sine wave generation
12.1.7T3 Features of an ac waveform
- Cycle
 - Frequency
 - Period
 - Amplitude
- 12.1.7T4 Effects of R – L – C on voltage and current in ac circuit
- Series circuits
 - Parallel
- 12.1.7T5 Experiment on power factor

Suggested Learning/Teaching Resources

- Resistors
- Capacitors
- Inductors

12.1.8 SINGLE PHASE TRANSFORMERS

Theory

- 12.1.8T1 *Specific Objectives*
By the end of the sub module unit, the trainee should be able to:
- describe the construction of single phase transformers
 - explain the principle of operation of single phase transformers
 - test single phase transformers
 - state applications of single phase transformers

12.1.7C Competence

The trainee should have the ability to:

- operate transforms
- connect transformers to single phase and three phase circuits
- connect for operations of an instrument transformers

Content

- 12.1.8T1 Construction of single

phase transformer
12.1.8T2 Explanation of
operation of single
phase transformers

- i) Transformer equation
- ii) Voltage transformation
- iii) Current transformation
- iv) Impedance transformation
- v) Equivalent circuit
- vi) Phasor diagram
- vii) Transformer efficiency

12.1.8T4 Testing of transformers

- i) No load test
- ii) Short circuit
- iii) Transformer polarity test

12.1.8T5 State applications
of single phase
transformers

- i) Matching
- ii) Auto-transformer

Practice

12.1.8P0 *Specific Objectives*

By the end of the sub
module unit, the trainee
should be able to:

- i) test transformers for proper operation
- ii) connect transformers to single phase and three phase circuits
- iii) connect instrument transformers

Content

12.1.8P1 Transformer tests

- i) Connectivity
- ii) Earthing
- iii) Insulation resistance
- iv) Open circuit
- v) Short circuit
- vi) Efficiency

12.1.8P2 Connection of
transformers

- i) Schematic diagrams
- ii) Wiring diagrams

12.1.8T3 Instrument transformers

Suggested teaching/Learning Activities

- Discussion
- Illustration
- Demonstration
- Note taking
- Practical exercise
- Calculations

Suggested teaching/Learning Resources

- Electrical measuring instruments
- Electronic tool kit
- Electrical tool kit
- Ac power supply
- Dc Power supply

Suggested Evaluation Methods

- Oral tests
- Timed written tests
- Assignments
- Timed practical tests
- Project